Sample Trunk Joint-Angle Data for Joshlene

This folder contains a small sample of trunk joint-angle data from the RE-MOVE Phase 1 pipeline to help you get familiar with the data format before working with the full dataset.

The sample includes one participant (P01), four trials, and two segment models.

Folder Structure

```
None
Sample Joint Angle Data For Joshlene/
├── 1 Segment Model/
│   ├── P01_20260806_NPose_T01/
│   ├── P01_20260806_TrunkFlexExt_T01/
│   ├── P01_20260806_TrunkRot_T01/
│   └── P01_20260806_LatBend_T01/
└── 2 Segment Model/
    └── Same four trials
```

Each trial folder contains an Excel file with the joint-angle data and a PNG file showing the corresponding plots.

Trials

Trial	Description	Approx. Duration
`NPose`	Quiet standing in the neutral calibration posture	13 s
`TrunkFlexExt`	Forward and backward trunk bending	55 s
`TrunkRot`	Trunk axial rotation	50 s
`LatBend`	Side-to-side trunk bending	48 s

The `NPose` trial can be used as a quick check of the data, since the angles should remain close to 0° in the calibration posture.

Segment Models

1 Segment Model

The trunk is treated as one segment relative to the pelvis. Only one joint, `Trunk`, is reported.

2 Segment Model

The trunk is divided into the lumbar and thorax segments. The following joint angles are reported:

- `Lumbar`: Lumbar segment relative to the pelvis.
- `Thorax`: Thorax relative to the pelvis, representing the overall upper-trunk motion.
- `ThoraxOnLumbar`: Thorax relative to the lumbar segment.

The full pipeline also includes 3-, 4-, and 6-segment models, which are not included in this sample.

Excel File Format

Each Excel file contains the following columns:

Column	Description
`Frame`	Frame number in the original motion-capture recording.
`Time_s`	Time in seconds from the start of the trial.
`*_Flex_Ext_deg`	Flexion (+) and extension (–), in degrees.
`*_LatBend_deg`	Right (+) and left (–) lateral bending, in degrees.
`*_AxRot_deg`	Right (+) and left (–) axial rotation, in degrees.

The `*` represents the joint name: `Trunk` for the 1-segment model, or `Lumbar`, `Thorax`, and `ThoraxOnLumbar` for the 2-segment model.

The data are sampled at 100 Hz, corresponding to one sample every 0.01 seconds.

Angle Calculation

All angles are calculated relative to the participant's standing calibration trial. The offset-removal method used for this sample is dRSM.

A value of 0° therefore represents the participant's calibration posture, rather than an absolute anatomical reference.

PNG Plots

Each PNG file corresponds to the Excel file in the same folder. The plots show joint angles over time, with separate plots for flexion/extension, lateral bending, and axial rotation.